Process FMEA - Bracket Welding Cell

Manufacturing Knowledge Hub | Process Engineering

Owner: Carlos Ibarra | Last updated: October 30, 2026

Failure mode and effects analysis for the automated welding cell, ranking failure modes by risk priority number for mitigation planning.

Key Points

- Highest RPN failure mode is inconsistent weld penetration from electrode wear.
- Mitigation adds an automated electrode wear check every 500 cycles.
- Detection controls were upgraded from manual visual check to inline vision inspection.
- FMEA is reviewed and updated after any process or equipment change.

Source: Manufacturing Knowledge Hub knowledge base | SharePoint Online